

Topic to discuss

- Finite difference Method
- Numerical Problem
- Homework Problem

Finite Difference Method :

It is a numerical Method used to solve differential equations by replacing derivatives with finite difference formula.

It can be used for first-order and second-order differential equations, but in Numerical Methods syllabus, it is mostly used for second-order boundary value problems.

For first order derivative, we can use three formulas.

Forward difference formula:

$$\frac{dy}{dx} \approx \frac{y_{i+1} - y_i}{h}$$

Backward difference formula:

$$\frac{dy}{dx} \approx \frac{y_i - y_{i-1}}{h}$$

Central difference formula:

$$\frac{dy}{dx} \approx \frac{y_{i+1} - y_{i-1}}{2h}$$

Note: for boundary value problems, central difference formula is mostly used for first derivative because it gives better accuracy.

For second-order derivative, we use:

$$\frac{d^2y}{dx^2} \approx \frac{y_{i+1} - 2y_i + y_{i-1}}{h^2}$$

In boundary value problems, values are given at two boundary points, such as:

$$y(a) = \alpha, \quad y(b) = \beta$$

The unknown values between the boundary points are calculated using finite difference equations.

Numerical Problem :

Q: Using finite difference method, solve the boundary value problem $\frac{d^2y}{dx^2} + y = 0$ with $y(0)=0$, $y(1)=1$

By taking $h=0.25$

Solution: Given, $h=0.25$, $\frac{d^2y}{dx^2} + y = 0$

$$x_0 = 0$$

$$x_1 = x_0 + h = 0.25$$

$$x_2 = x_1 + h = 0.50$$

$$x_3 = x_2 + h = 0.75$$

$$x_4 = x_3 + h = 1$$

We have boundary value, $y_0=0$, $y_4=1$

We need to find y_1, y_2, y_3
We know from finite difference formula,

$$\frac{d^2y}{dx^2} \approx \frac{y_{i+1} - 2y_i + y_{i-1}}{h^2}$$

Now substitute the equation, we get

$$\frac{d^2y}{dx^2} + y_i = 0$$

$$\Rightarrow \frac{y_{i+1} - 2y_i + y_{i-1}}{h^2} + y_i = 0$$

$$\left[\begin{array}{l} \because h = 0.25 \\ h^2 = 0.0625 \end{array} \right]$$

$$\Rightarrow \frac{y_{i+1} - 2y_i + y_{i-1}}{0.0625} + y_i = 0$$

$$\Rightarrow \frac{y_{i+1} - 2y_i + y_{i-1}}{0.0625} + y_i = 0$$

$$\Rightarrow y_{i+1} - 2y_i + y_{i-1} + 0.0625 y_i = 0$$

$$\Rightarrow y_{i+1} - 1.9375 y_i + y_{i-1} = 0 \text{ ————— } \textcircled{1}$$

Applying formula at interior point,
interior points are x_1, x_2, x_3

So, $i = 1, 2, 3$

For $i=1$,

$$y_{i+1} - 1.9375 y_i + y_{i-1} = 0$$

$$\Rightarrow y_2 - 1.9375 y_1 + y_0 = 0$$

$$\Rightarrow y_2 - 1.9375 y_1 = 0$$

$$\Rightarrow y_2 = 1.9375 y_1 \quad \text{--- (2)}$$

for $i=2$,

$$y_3 - 1.9375 y_2 + y_1 = 0$$

$$\Rightarrow y_3 - 1.9375 (1.9375 y_1) + y_1 = 0$$

$$\Rightarrow y_3 = 2.7539062 y_1 \quad \text{--- (3)}$$

for $i = 3$,

$$y_{i+1} - 1.9375 y_i + y_{i-1} = 0$$

$$\Rightarrow y_4 - 1.9375 y_3 + y_2 = 0$$

$$\Rightarrow 1 - 1.9375(2.7539062 y_1) + 1.9375 y_1 = 0 \quad \left[\begin{array}{l} \text{using eq. 2} \\ \& \text{eq. 3} \end{array} \right]$$

$$\Rightarrow 1 - 3.3981933 y_1 = 0$$

$$\therefore y_1 = \frac{1}{3.3981933}$$

$$\therefore y_1 = 0.2942740$$

$$\therefore y_2 = 1.9375 y_1 = 0.5701559 \quad [\text{from eqn. -2}]$$

$$y_3 = 2.7539062 y_1 = 0.810402 \quad [\text{from eqn. -3}]$$

final answers,

$$x_0 = 0$$

$$x_1 = 0.25$$

$$x_2 = 0.50$$

$$x_3 = 0.75$$

$$x_4 = 1$$

$$, y_0 = 0$$

$$, y_1 = 0.2942740$$

$$, y_2 = 0.5701559$$

$$, y_3 = 0.810402$$

$$, y_4 = 1$$

Ans.

Homework Problem :

Q: Using finite difference method solve the boundary value problem,

$$\frac{d^2 y}{dx^2} + y + 1 = 0 \quad \text{with} \quad y(0) = 0, \quad y(1) = 0$$

Solution : Given,

$$\frac{d^2 y}{dx^2} + y + 1 = 0$$

$$\text{Let, } h = 0.25$$

$$\text{So, } x_0 = 0, \quad x_1 = 0.25, \quad x_2 = 0.50, \quad x_3 = 0.75, \quad x_4 = 1$$

$$\text{and } y_0 = 0, \quad y_4 = 0$$

Applying finite difference equation of 2nd order,
we get,

$$\frac{d^2 y}{dx^2} + y + 1 = 0$$

$$\Rightarrow \frac{y_{i+1} - 2y_i + y_{i-1}}{h^2} + y_i + 1 = 0 \quad \left[\begin{array}{l} h = 0.25 \\ \therefore h^2 = 0.0625 \end{array} \right]$$

$$\Rightarrow y_{i+1} - 2y_i + y_{i-1} + 0.0625 y_i + 0.0625 = 0$$

$$\Rightarrow y_{i+1} - 1.9375 y_i + y_{i-1} + 0.0625 = 0$$

We need to find, y_1, y_2 & y_3

So, $i = 1, 2, 3$

for $i=1$,

$$y_{i+1} - 1.9375 y_i + y_{i-1} + 0.0625 = 0$$

$$\Rightarrow y_2 - 1.9375 y_1 + y_0 + 0.0625 = 0 \quad [\because y_0 = 0]$$

$$\Rightarrow y_2 = 1.9375 y_1 - 0.0625 \quad \text{--- (1)}$$

for $i=2$,

$$y_3 - 1.9375 y_2 + y_1 + 0.0625 = 0$$

$$\Rightarrow y_3 - 1.9375 (1.9375 y_1 - 0.0625) + y_1 + 0.0625 = 0$$

$$\Rightarrow y_3 = 2.753906 y_1 - 0.183593 \quad \text{--- (2)}$$

for $i=3$,

$$y_{i+1} - 1.9375 y_i + y_{i-1} + 0.0625 = 0$$

$$y_4 - 1.9375 y_3 + y_2 + 0.0625 = 0 \quad [\because y_4 = 0]$$

$$\Rightarrow 0 - 1.9375 (2.753906 y_1 - 0.183593) + (1.9375 y_1 - 0.0625) + 0.0625 = 0$$

$$\Rightarrow -3.3981928 y_1 + 0.3557114 = 0$$

$$\therefore y_1 = 0.104676$$

Now using -①

$$y_2 = 1.9375 (0.104676) - 0.0625 \\ = 0.140312$$

Now using —(2)

$$y_3 = 2.753906 (0.104676) - 0.183593$$
$$= 0.104677$$

So, $y_1 = 0.104676$

$$y_2 = 0.140312$$

$$y_3 = 0.104677$$

Ans.

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